§4 TIPA, PSL and ISL — part 3 of 4

Reference: subsec:psl, subsec:purpose_action_result

Source: paper/section4_tipa.tex, lines 1684–2349 (the Physical Substrate Layer PSL, including PSL-1 to PSL-4, the resolution of the M3 action command and the local protective response, PSL sensitivity control and the minimal structural condition; and the description of purpose, action and result), based on commit e6e12cd, with the author's corrections to lines 2340 and 2342 recorded in ledger entry L-A3.187. The Japanese manuscript is authoritative. Part 3 of §4 follows 04_tipa_2.md. Part 4 (ISL, integrated high-speed processing and Automaticity, the five functional memory sets, the functional axis, and the closing summary, lines 2350–2964) is in 04_tipa_4.md.

PSL: the Physical Substrate Layer

Reference: subsec:psl

PSL is a classification-B standard functional form that functionally connects, between TIPA and ISL, input and output, memory I/O, substrate maintenance, the local protective response, sensitivity control and the formation of the effective action. PSL is not a fourth cognitive mechanism and does not independently re-execute the cognitive processing of M1, M2 and M3.

Definition (def:psl_availability): availability (classification C). Information, a signal, a resource or an action capability is *available* to a mechanism when it is in a state accessible as a permitted input or object of action of that mechanism. This paper does not state, from availability, actual use, correct interpretation or the success of later processing as a general consequence.

Definition (def:psl): formal definition of PSL (classification B). PSL is fixed as the standard functional configuration that bears input acquisition, memory I/O, realization of action, substrate maintenance, the local protective response, sensitivity control and the resolution of the effective action. The concrete internal arrangement, physical medium, communication method and implementation topology are not fixed.

PSL-1: input and memory-output interface

Reference: subsubsec:psl1

PSL-1 connects external and internal signals on the implementation side, and memory output, to information available to TIPA.

Definition (def:psl1): PSL-1 (classification B). With $\mathcal{S}_{\mathrm{in}}$ the implementation-side input signal space and an input interface $\Phi_{\mathrm{sense}} : \mathcal{S}_{\mathrm{in}} \to \mathcal{I}$, put $I_{\mathrm{raw},t} := \Phi_{\mathrm{sense}}(s_{\mathrm{in},t})$. When input sensitivity control is used, $I_{\mathrm{sense},t} := \Gamma_{\mathrm{in},t}(I_{\mathrm{raw},t}) \in \mathcal{I}$. With $\mathcal{S}_{\mathrm{mem,out}}$ the output signal space on the memory-implementation side and $\mathcal{I}_{\mathrm{mem,out}}$ the space of memory-output information available to TIPA, a memory-output interface $\Phi_{\mathrm{mem_out}} : \mathcal{S}_{\mathrm{mem,out}} \to \mathcal{I}_{\mathrm{mem,out}}$ is given.

Role separation of PSL-1 and M1 (rem:psl1_m1). PSL-1 bears the connection that makes information available. It does not estimate the state or the environment, discriminate, or evaluate feasibility-related quantities.

PSL-2: output and memory-input interface

Reference: subsubsec:psl2

PSL-2 connects the effective action determined by PSL to the acting-side implementation and, as needed, makes information to be retained enterable on the memory-implementation side.

Definition (`def:psl2`): PSL-2 (classification B). With a non-empty drive-signal space $\mathcal{S}_{\text{drive}}$ and a map $\Phi_{\text{action}} : U \to \mathcal{S}_{\text{drive}}$, put $x_{\text{drive,raw},t} := \Phi_{\text{action}}(u_t)$ for the effective action $u_t \in U$. When output sensitivity control is used, $x_{\text{drive},t} := \Gamma_{\text{out},t}(x_{\text{drive,raw},t})$. With $\mathcal{I}_{\text{mem,in}}$ the information space handed from TIPA to the memory-implementation side and $\mathcal{S}_{\text{mem,in}}$ the input signal space of that side, a memory-input interface $\Phi_{\text{mem_in}} : \mathcal{I}_{\text{mem,in}} \to \mathcal{S}_{\text{mem,in}}$ is given.

Boundary of memory input and Update (`rem:mem_input_update_boundary`). PSL-2 can connect information to the memory-implementation side but does not fix the $t \to t+1$ update rule of the retained content itself. That dynamics is the responsibility of §5 (`sec:extended_dynamics`).

PSL-3: substrate maintenance and substrate state signals

`Reference: subsubsec:psl3`

PSL-3 maintains the substrate and resource conditions under which TIPA, PSL and ISL can operate, and makes their state available to the mechanisms that need it.

Definition (`def:psl3`): PSL-3 (classification B). The PSL substrate state is $s_{\text{PSL},t} \in \mathcal{S}_{\text{PSL}}$. With $\mathcal{R}_{\text{PSL}}$ the space of available resources, the substrate-maintenance map is $\Phi_{\text{maint}} : \mathcal{S}_{\text{PSL}} \times \mathcal{R}_{\text{PSL}} \to \mathcal{S}_{\text{PSL}}$. Substrate information that can become available to TIPA may be extracted by $\Phi_{\text{base}} : \mathcal{S}_{\text{PSL}} \to \mathcal{I}_{\text{base}}$.

Distinction of PSL-3 and the time-indexed satisfaction predicate of SM (`rem:psl3_sm_base`). PSL-3 relates to the substrate conditions involved in self-maintenance, but this paper does not state, from the existence or operation of PSL-3, the truth of the time-indexed satisfaction predicate of SM as a general consequence.

PSL-4: protective response to a local acute abnormality

`Reference: subsubsec:psl4`

PSL-4 is a classification-B standard functional form that can connect locally acquirable acute-abnormality information to a protective response without making completion of the detailed processing of M1, M2 and M3 a general requirement. “Acute” here does not mean a universal time threshold, deadline or maximum allowed delay.

Definition (`def:psl4`): PSL-4, protective response to a local acute abnormality (classification B). With a local signal space $\mathcal{S}_{\text{local}}$ and the set of conditions requiring a local protective response $\mathcal{C}_{\text{prot}} \subseteq \mathcal{S}_{\text{local}}$, the local-protective-response generating map is $\Phi_{\text{prot}} : \mathcal{S}_{\text{local}} \to U \cup \{\bot\}$ and $r_{\text{local},t} := \Phi_{\text{prot}}(s_{\text{local},t}) \in U \cup \{\bot\}$. $\bot$ means that no local protective response is generated. As a standard operating condition, $s_{\text{local},t} \notin \mathcal{C}_{\text{prot}} \Rightarrow \Phi_{\text{prot}}(s_{\text{local},t}) = \bot$. The concrete content of $\mathcal{C}_{\text{prot}}$, the composition of local signals, the kinds of response and the time scale depend on the implementation.

Acuteness, locality and non-deliberation of PSL-4 (`rem:psl4_no_deliberation`). “Acute” expresses the functional division that a protective response can be connected without making completion of the detailed TIPA processing a general requirement; it does not mean a universal time threshold. “Local” expresses a functional connection between local abnormality information and a protective response, not the physical location of a particular organ.

Structure versus successful operation of PSL-4 (`rem:psl4_required`). Having the standard structure of PSL-4 is not the same as an appropriate protective response being generated at each time. This paper does not state as a general consequence, from $r_{\text{local},t} \neq \bot$, the truth of the time-indexed satisfaction predicate of SM, avoidance of blockage or avoidance of collapse. Item (4) of assumption `ass:psl_minimal` requires only the existence of a local signal generating a local protective response, and does not require that a protective response be generated at each time.

Resolution of the M3 action command and the local protective response

Reference: subsubsec:effective_action_resolution

The $u_{\mathrm{M3},t}$ formed by M3 is a TIPA-side action command and can in general differ from the final effective action u_t entering the state transition. PSL determines u_t by deterministically resolving the relation to the local protective response and other inputs.

Definition (def:effective_action_resolve): effective-action resolution map (classification B). With an auxiliary-information space $\mathcal{R}$ for resolution, the map is $\Phi_{\mathrm{resolve}} : U \times (U \cup \{\bot\}) \times \mathcal{R} \to U$, and $u_t = \Phi_{\mathrm{resolve}}(u_{\mathrm{M3},t}, r_{\mathrm{local},t}, R_t) \in U$. When there is no local protective response, $\Phi_{\mathrm{resolve}}(u_{\mathrm{M3},t}, \bot, R_t) = u_{\mathrm{M3},t}$ is the standard degeneration condition.

A fixed map $\mathrm{pr}_{\mathrm{loc}} : U \to U_{\mathrm{loc}}$ is placed, with U_{loc} non-empty; $\mathrm{pr}_{\mathrm{loc}}(u)$ is the *local-response component* of u. The standard form places two conditions:

- (a) *Non-degeneracy of the local protective response:* for every $s \in \mathcal{S}_{\mathrm{local}}$ with $\Phi_{\mathrm{prot}}(s) \neq \bot$, $\mathrm{pr}_{\mathrm{loc}}(\Phi_{\mathrm{prot}}(s)) \neq \mathrm{pr}_{\mathrm{loc}}(u_\emptyset)$.
- (b) *Preservation of firing:* for every $u \in U$, $r \in U$, $R \in \mathcal{R}$, $\mathrm{pr}_{\mathrm{loc}}(\Phi_{\mathrm{resolve}}(u, r, R)) = \mathrm{pr}_{\mathrm{loc}}(r)$.

(b) is satisfied by a resolution with $\Phi_{\mathrm{resolve}}(u, r, R) = r$, so it is a non-vacuous condition. By (a) the local-response component of a local protective response differs from that of selective waiting, and (b) requires that it not be replaced by waiting or another value through the M3 action command. The concrete contents of U_{loc} and $\mathrm{pr}_{\mathrm{loc}}$ depend on the implementation. The local-response component is an abstract component identifying the firing of a local protective response on the effective action; it does not express a particular degree of freedom of a body, a waveform of movement or its strength. What (b) requires is that the firing is not erased from the effective action by the M3 action command. That the strength or shape of a fired response changes through interference with other actions can be expressed as a difference in components other than the local-response component, and (b) does not exclude it.

At a time when a local protective response occurs, the local-response component of the effective action takes the value of the local protective response (preservation of firing). Whether the remaining components follow the M3 action command, or are overwritten, modified, composed or otherwise resolved, depends on the implementation. This paper does not assume that the local protective response replaces the whole effective action.

Local protective response and the concurrency with TIPA (rem:psl4_concurrency). The local protective response $r_{\mathrm{local},t} = \Phi_{\mathrm{prot}}(s_{\mathrm{local},t})$ is generated only from the local signal $s_{\mathrm{local},t} = \rho_{\mathrm{local}}(I_t)$ (def:s4_psl_sources) and does not use outputs of TIPA. It can therefore occur at the same time as, and separately from, TIPA processing. That the resolution map takes both $u_{\mathrm{M3},t}$ and $r_{\mathrm{local},t}$ of the same time as arguments expresses the formal combining of two values into the effective action of that time; it does not make completion of M3 processing a condition for the occurrence of a local protective response. This paper does not fix a physical precedence relation within one time.

The firing of a local protective response is not cancelled by the M3 action command (preservation of firing in def:effective_action_resolve). If a local protective response occurs during the execution of an action, the local-response component follows the local protective response and the other components can follow the M3 action command. A local protective response can therefore occur in parallel during the execution of an action.

The influence after a local protective response is integrated by M1 at the next time and later, through the record u_{t-1} of the final effective action and through changes of state and sensing (rem:tipa_bidirectional). Since M3 does not take the local protective response of the same time as input, an M3 action responding to it is chosen at the next time or later. A response by which M3 stops a local protective response is not the cancellation of firing but a response after firing. It is expressed as follows: the action's effect reaches the local signal of a later time through the state, the local signal ceases to enter the condition set $\mathcal{C}_{\mathrm{prot}}$, and a local protective response is therefore no longer generated at the later time. If a local protective response is generated again at a later time, its firing is preserved in the same way. At a time when the

local signal does not enter $\mathcal{C}_{\text{prot}}$, no local protective response has fired and preservation of firing has no object. Making the local signal not enter $\mathcal{C}_{\text{prot}}$ through an earlier action (for example, avoiding a stimulus beforehand) is not stopping the firing but keeping the condition of firing from being met. This paper distinguishes keeping the condition of firing from being met, stopping the continuation after firing, and preventing firing again. For example, when a local protective response such as coughing or blinking is expressed in the form of PSL-4, what M3 can choose is not to stop the firing itself but a response that stops its continuation after firing.

PSL sensitivity control

`Reference: subsubsec:psl_sensitivity`

PSL sensitivity control adjusts the connection characteristics on the input side or the output side according to control information already available.

Definition (`def:psl_sensitivity`): PSL sensitivity-control map (classification B). With families $\mathfrak{G}_{\text{in}}, \mathfrak{G}_{\text{out}}$ of permitted sensitivity maps on the input and output sides, the sensitivity maps at time t are $\Gamma_{\text{in},t} \in \mathfrak{G}_{\text{in}}$ and $\Gamma_{\text{out},t} \in \mathfrak{G}_{\text{out}}$, acting in a type-consistent way at least as $\Gamma_{\text{in},t} : \mathcal{J} \to \mathcal{J}$ and $\Gamma_{\text{out},t} : \mathcal{S}_{\text{drive}} \to \mathcal{S}_{\text{drive}}$. Parameterization, selection rules, continuity and inter-time update rules are not fixed. However, $\Gamma_{\text{in},t}$ is determined before the TIPA processing of that time that uses $I_{\text{sense},t}$. If internal TIPA-derived information is used to choose $\Gamma_{\text{in},t}$, it depends only on information already available before that TIPA processing at time t (in the standard form, TIPA outputs up to the previous time), and a same-time circularity that re-determines $\Gamma_{\text{in},t}$ from the outputs of M1, M2 and M3 within the same step is not allowed.

Definition (`def:s4_psl_sources`): signal and substrate carriers of PSL and their connection to the standard maps (classification B). The following readouts are fixed: $\rho_{\text{in}} : X \times E_{\text{ext}} \to \mathcal{S}_{\text{in}}$, $\rho_{\text{PSL}} : X \to \mathcal{S}_{\text{PSL}}$, $\rho_{\Gamma,\text{in}} : X \to \mathfrak{G}_{\text{in}}$, $\rho_{\Gamma,\text{out}} : X \to \mathfrak{G}_{\text{out}}$, $\rho_{\text{mem,out}} : X \to \mathcal{S}_{\text{mem,out}}$, $\rho_{\text{local}} : \mathcal{J} \to \mathcal{S}_{\text{local}}$. Then $s_{\text{in},t} = \rho_{\text{in}}(x_t, e_t)$, $s_{\text{PSL},t} = \rho_{\text{PSL}}(x_t)$, $\Gamma_{\text{in},t} = \rho_{\Gamma,\text{in}}(x_t)$ and $\Gamma_{\text{out},t} = \rho_{\Gamma,\text{out}}(x_t)$. The connection of PSL-1 and standard sensing is specified so that $P_{\text{sens}}(x, e) = \rho_{\Gamma,\text{in}}(x)(\Phi_{\text{sense}}(\rho_{\text{in}}(x, e)))$ holds. This equality is not a theorem derived from arbitrary existing maps; it is a connection condition when the standard form of this subsection is adopted. In a realization that uses the memory-implementation output $\Phi_{\text{mem_out}}(\rho_{\text{mem,out}}(x_t))$ and the substrate information $\Phi_{\text{base}}(\rho_{\text{PSL}}(x_t))$ as components of the sensed information, fixed readouts $\mathcal{J} \to \mathcal{J}_{\text{mem,out}}$ and $\mathcal{J} \to \mathcal{J}_{\text{base}}$ are connected so that they return these values. This is a condition on signals of the physical carrier and not a requirement to recover an arbitrary M_t from x_t; functionally retained content is available separately from M_t. The local signal is $s_{\text{local},t} = \rho_{\text{local}}(I_t)$. For the auxiliary information of resolution a fixed map $\Phi_R : \mathcal{M} \times \mathcal{J} \to \mathcal{R}$ is used, with $R_t = \Phi_R(M_t, I_t)$; the effective action is therefore not changed by an unobserved difference of state alone under the same M, I.

A map $\rho_{\text{res}} : X \times U \times E_{\text{ext}} \to \mathcal{R}_{\text{PSL}}$ reading the resource input to substrate maintenance is fixed. PSL-3 is not made an independent state transition; in the adopted realization it is made consistent with f by $\rho_{\text{PSL}}(f(x, u, e)) = \Phi_{\text{maint}}(\rho_{\text{PSL}}(x), \rho_{\text{res}}(x, u, e))$. The actual action that depends on the drive signal of PSL-2 and the state of the effectors is also included in f, and no independent inter-time state is placed on the drive signal. The successor value of the sensitivity is $\rho_{\Gamma,\cdot}(f(x_t, u_t, e_t))$; it can include change through directed internal action but is not limited to it.

When memory input is used, a fixed map $\Phi_{\text{write}} : \mathcal{M} \times \mathcal{J} \to \mathcal{J}_{\text{mem,in}}$ gives the information to be written, which $\Phi_{\text{mem_in}}$ connects to implementation signals. The next state of the retained content is determined by Update, and no separate memory-update rule is placed alongside. The correspondence between write signals and Update in an implementation is specified within the range of the types of the adopted component updates.

Non-cognitive character of sensitivity control (`rem:sensitivity_noncognitive`). PSL sensitivity control is not cognitive evaluation, Priority generation, judgment of the operating mode, PSL-4, Automaticity or integrated high-speed processing itself.

Minimal structural condition of PSL

Reference: subsubsec:psl_minimal_conditions

Assumption (ass:psl_minimal): minimal structural condition and non-cognitive character of PSL. The PSL of an ACS treated in this subsection satisfies at least the following.

1. PSL-1 can make the necessary input and memory output available to TIPA.
2. PSL-2 can connect the final effective action and necessary memory input to the implementation side.
3. PSL-3 can bear the necessary substrate and resource-maintenance functions.
4. PSL-4 can carry a protective response to a local acute abnormality; that is, $\Phi_{\mathrm{prot}}(s) \neq \bot$ for some $s \in \mathcal{C}_{\mathrm{prot}}$.
5. PSL does not independently substitute for the cognitive responsibilities of M1, M2 and M3.

The concrete physical parts, circuits, communication methods and software configuration are not fixed. *Introduction status: S (structural). Strength: medium.* The assumption limits the range of ACSs to which the standard form of PSL in subsec:psl applies, and is not used as a premise of a proposition of this subsection.

Description of purpose, action and result

Reference: subsec:purpose_action_result

Under the standard forms of subsec:tipa_architecture and subsec:psl, this subsection gives the attribution of purposes, the involvement of requests, the classification of actions and the description of the result of one action. None introduces a new mechanism or state variable.

Proposition (prop:purpose_self_attribution): self-attribution of purposes. Under the standard form of this section, Req_t, sig_t, $\mathrm{ev}_t^{\mathrm{rq}}$, Pur_t and stat_t at each time t are determined as values of fixed maps with (M_t, I_t) as arguments. Hence the criteria of requests or evaluations of a subject external to the ACS enter them only through the sensed information I_t or through the retained content M_t.

Proof (summary). By def:s4_carrier_contract the time-dependent mechanism outputs are generated under fixed rules from (M_t, I_t), and previous-time symbols are values of fixed readouts of the component M_t^{W}. $\mathrm{Req}_t = \rho_{\mathrm{req}}(M_t)$ (def:request_signal_types). The values of M1 are determined by $I_{\mathrm{sense},t} = I_t$, M_t and previous-time symbols (def:m1_map), those of M2 by M1's outputs, M_t and previous-time symbols (def:m2_map), so sig_t and $\mathrm{ev}_t^{\mathrm{rq}}$ are values of fixed maps of (M_t, I_t). The inputs of M3 consist of these, M_t and previous-time symbols (def:m3_map), so Pur_t and stat_t are likewise. The second half follows from the arguments of the maps being limited to (M_t, I_t). □

The proposition is a statement about values of fixed maps. Whether these time-indexed values are attributed to Σ_i as its outputs follows def:s4_output_attribution.

Scope of self-attribution and abnormal ranges (rem:self_attribution_scope). The proposition is a structural property that the criteria of purposes are read out from carriers inside the ACS; it does not mean that purposes aim at the ACS's own benefit or at maintaining Π. A purpose aimed at the protection of others or at self-sacrifice also has this property as long as it is formed from requests the ACS holds. An external subject's direct determination of the action does not occur at the level of types, because the final effective action u_t is determined only as a value of Φ_{resolve} (def:effective_action_resolve). On the other hand, when M_t is rewritten by an external action, it cannot at the level of types be distinguished from a change by learning. Malfunction, manipulation and the bypassing or replacement of the evaluation mechanism by exogenous domination remain within the description of this remark; tracing their origin is a task from Phase 1.α on (subsec:phase1a).

Definition (`def:req_involvement`): involvement of requests (classification C). For time t, the inputs of M3 other than the triple of the list of requests, request signal and request-by-object evaluation (namely $\hat{x}_t$, $\hat{e}_t$, ξ_t, P_t, ev_t, Pur_{t-1}, M_t), together with $r_{\mathrm{local},t}$ and R_t of the effective-action resolution, are fixed. For a triple $(R, \gamma, \nu) \in \mathcal{P}_{\mathrm{fin}}(\mathrm{Req}) \times \mathrm{Sig} \times \mathfrak{C}^{\mathrm{rq}}_{\mathrm{map}}$, write $\mathcal{U}_t(R, \gamma, \nu) \in U$ for the value obtained by resolving, with Φ_{resolve}, the M3 action command that takes these as inputs. For a request $q \in \mathrm{Req}_t$ put $R_t^{\setminus q} := \mathrm{Req}_t \setminus \{q\}$, $\mathrm{sig}_t^{\setminus q} := \{(q', c) \in \mathrm{sig}_t \mid q' \neq q\}$ and $\nu_t^{\setminus q} := \mathrm{ev}_t^{\mathrm{rq}}$ restricted to $D_t^{\mathrm{rq}} \setminus (\{q\} \times (\mathrm{Lab} \cup \{*\}))$. If $\mathcal{U}_t(\mathrm{Req}_t, \mathrm{sig}_t, \mathrm{ev}_t^{\mathrm{rq}}) \neq \mathcal{U}_t(R_t^{\setminus q}, \mathrm{sig}_t^{\setminus q}, \nu_t^{\setminus q})$, the request q is said to have *been involved* in the effective action at time t; otherwise q is said to have existed at time t but not to have been involved. By `def:m3_map` the right-hand side is also determined.

Scope of the involvement judgment (`rem:involvement_scope`). The definition judges involvement through the explicit routes of the list of requests, the request signal and the request-by-object evaluation. Since the retained content M_t is fixed, influence through M_t is not included. A judgment of "involved" therefore states that the request affected the result on this map, whereas a judgment of "not involved" states only that it did not affect the result through the explicit routes, and does not mean there is no influence through M_t. For example, if a response originating in a request has been stored as a habit in the retained content and the same action occurs even when that request is removed from the request signal, the action has no involvement of that request through the explicit routes. The involvement judgment is made on the final effective action u_t; a similar judgment on $u_{\mathrm{M3},t}$ is treated as an auxiliary description.

Definition (`def:action_classes`): classification of actions (classification C). The effective action u_t at time t is classified as follows. (i) If some $q \in \mathrm{Req}_t$ was involved in u_t, u_t is called a *purposive action*. (ii) If no $q \in \mathrm{Req}_t$ was involved, u_t is called a *self-generated action not mediated by a purpose*. A case where a local protective response (`subsec:psl`) overwrote the M3 action command, and a response by Automaticity (`subsec:fast_processing`), may be classified under (ii). External forcing is not an action of the ACS concerned; it is expressed as a state change entering f through the external environment state e_t (`rem:action_space_structure` of §2). At the same time, a purposive action and external forcing can coexist.

Definition (`def:purpose_action_difference`): action difference relating to a purpose and one-step condition satisfaction (classification C). The attainment condition that the describer assigns to a purpose $p \in \mathrm{Pur}$ is $G_p \subseteq X \times O$; G_p is not an internal quantity of the ACS. For $e = (b, o)$ the joint transition of the state and the action-response component is $\bar{f}(x, u, e) := (f(x, u, e), g_O(o, x, u, b)) \in X \times O$ (`def:env_realization` of §2). If $\mathbf{1}[\bar{f}(x, u, e) \in G_p] \neq \mathbf{1}[\bar{f}(x, u_\emptyset, e) \in G_p]$, the action u is said to have an *action difference* relating to the purpose p at (x, e). That $(x_{t+1}, o_{t+1}) \in G_p$ is called the *one-step condition satisfaction* after the action at time t. This is not attainment of a purpose in general (`def:purpose_course`).

Proposition (`prop:purpose_ad_slot`): division on the external side and the action difference of a slot. For some $G_O \subseteq O$ let $G_p = X \times G_O$. If an action u has an action difference relating to the purpose p at (x, e), then for some slot $a \in \mathcal{A}_O$, u has an action difference on the slot a at (o, x, b) (`def:action_difference` of §2).

Proof (summary). Since $G_p = X \times G_O$, $\mathbf{1}[\bar{f}(x, u, e) \in G_p] = \mathbf{1}[g_O(o, x, u, b) \in G_O]$, and likewise for $u_\emptyset$. By hypothesis these two values differ, so $g_O(o, x, u, b) \neq g_O(o, x, u_\emptyset, b)$. Since $O = \prod_{a \in \mathcal{A}_O} O_a$, the a-th components differ for some a. If $\mathcal{A}_O = \emptyset$, O is a singleton and the hypothesis cannot hold. □

This paper does not state the converse of `prop:purpose_ad_slot` as a general consequence.

External forcing and the action difference (`rem:forcing_action_difference`). The ACS's own contribution at a time when external forcing is present is described by the action difference with e_t, including the forcing, held fixed. If any action yields the same next state and action-response component, this is expressed as there being no action difference relating to the purpose for any action, and no new concept is placed. Forcing that actually occurred is described on the trajectory as e_t, and the admissible external action $\Upsilon_{\mathrm{adm},i}$ is used only for the judgment of recoverability (`def:collapse` of §2). The action difference is a one-step quantity; the comparison of contributions over several steps, and the whole trajectory in the case

where no action was taken, are not fixed in this paper.

Definition (`def:result_tuple`): description of the result of one action (classification C). For a time t with $\mathrm{WF}_{i,t} = 1$ and $p \in \mathrm{Pur}_t$, the one-step result of the effective action u_t at time t with respect to the purpose p is described by the following triple.

1. *Action difference relating to the purpose:* whether u_t has an action difference relating to p at (x_t, e_t).
2. *One-step condition satisfaction:* whether $(x_{t+1}, o_{t+1}) \in G_p$. If $G_p = X \times G_O$, it is determined by o_{t+1} alone.
3. *Membership in the prediction set:* the truth or falsity of each of the internal deviation from prediction and the theoretical deviation from prediction (`def:prediction_record_check`). Each notion is undefined when its respective reference prediction is $\perp$. The internal deviation is also undefined when $\mathrm{WF}_{i,t+1} = 0$.

An undefined component is not treated as false. The internal judgment of attainment is not included in the one-step result and is treated in the description of the course of a purpose (`def:purpose_course`).

Definition (`def:purpose_course`): description of the course of a purpose (classification C). Suppose a purpose p newly enters the set of purposes at time τ ($p \in \mathrm{Pur}_\tau$ and, when $\tau \geq 1$, $p \notin \mathrm{Pur}_{\tau-1}$); here $\mathrm{Pur}_{\tau-1}$ is the output of M3 at time $\tau - 1$ and not a retained-record value. For the course of p starting from τ, the following times are defined; where no such time exists, it is not defined.

1. *Theoretical attainment time:* $s^{\mathrm{th}} := \min\{s > \tau \mid (x_s, o_s) \in G_p\}$.
2. *Internal attainment-judgment time:* $s^{\mathrm{in}} := \min\{s > \tau \mid \mathrm{stat}_s(p) = \mathsf{ach}\}$, where s ranges over times with $\mathrm{WF}_{i,s} = 1$ at which $\mathrm{stat}_s(p)$ is determined.
3. *End time:* $s^{\mathrm{end}} := \min\{s > \tau \mid \mathrm{WF}_{i,s} = 1,\ p \notin \mathrm{Pur}_s\}$, together with the *kind of ending* by the judgment at that time: ending by a judgment of attainment if $\mathrm{stat}_{s^{\mathrm{end}}}(p) = \mathsf{ach}$, ending without attaining if term, and disappearance without a judgment if und or not determined.

s^{th} is a quantity at the describer's level and s^{in} a quantity at the ACS's internal level; they need not coincide. For example $s^{\mathrm{th}} < s^{\mathrm{in}}$ can express realizing an attainment late, and the case where s^{in} is determined while s^{th} is not can express a mistaken judgment of attainment. A purpose that ends in one step is the special case $s^{\mathrm{th}} = \tau + 1$.

End of persistence and the description of the result (`rem:result_after_collapse`). For example, consider a parent who gave the last food to a child and later collapses. If the child's state is described as an action-response component seen from the parent (`rem:exogenous_environment` of §2), the action difference of the giving action is determined by the value at the time of the parent's action, while the condition satisfaction on the child's side and the theoretical attainment time are determined by the values of the child's component. The judgment of the state of the parent's purpose is not determined at times with $\mathrm{WF}_{i,t} = 0$ (`def:s4_output_attribution`), and is read neither as "ended without attaining" nor as "judgment withheld". A time that is not a collapse but has $\mathrm{WF}_{i,t} = 0$ (for example, a time at which functions have stopped and the mechanisms of this section are not realized) is treated in the same way. This paper does not state that the parent would have persisted had it not given. The self-attribution of purposes (`prop:purpose_self_attribution`) holds also for such a purpose aimed at self-sacrifice.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.